

Are All SNAP Dollars Created Equal? Purchasing Power Across Urban and Rural Counties

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December 2025

- SNAP is the largest food assistance program in the U.S., functioning as an automatic stabilizer.
- Existing literature focuses on:
 - SNAP's countercyclical enrollment during recessions.
 - Its role in stabilizing low-income households.
- Much less is known about **geographic variation in purchasing power**.
- Key question: **Does a SNAP dollar buy less in urban counties than rural counties?**

Motivation: Rural vs Urban Context

- Rural and urban areas differ sharply in:
 - Cost of living
 - Housing markets
 - Food access
- USDA finds higher food insecurity in rural counties, but higher cost-of-living in urban counties.
- If nominal benefits are fixed nationally, **real value may differ systematically across space.**

How does the **real purchasing power** of SNAP benefits vary with a county's level of urbanization?

- SNAP issuance data (USDA, 2010)
- Urban/rural population data (NHGIS 2010)
- Median home value (NHGIS 2010)
- Demographic controls (ACS 2010):
 - Race composition
 - Poverty rate
 - Educational attainment
 - Median income
 - Employment/unemployment
- Sample restricted to counties with population $\geq 65,000$

Constructing Real SNAP

- Step 1: Compute SNAP per capita for each county.
- Step 2: Adjust for cost of living using county median home value.
- Real SNAP per capita:

$$\text{RealSNAP}_i = \frac{\text{SNAP per capita}_i}{\text{Median Home Value}_i}$$

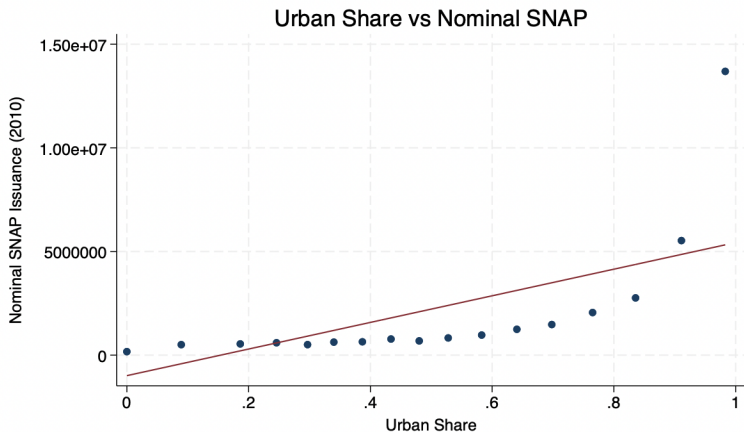
- Interpreted as a **relative purchasing power index**.
- Higher values = SNAP dollars stretch further.

Summary Statistics

Table 1: Summary Statistics

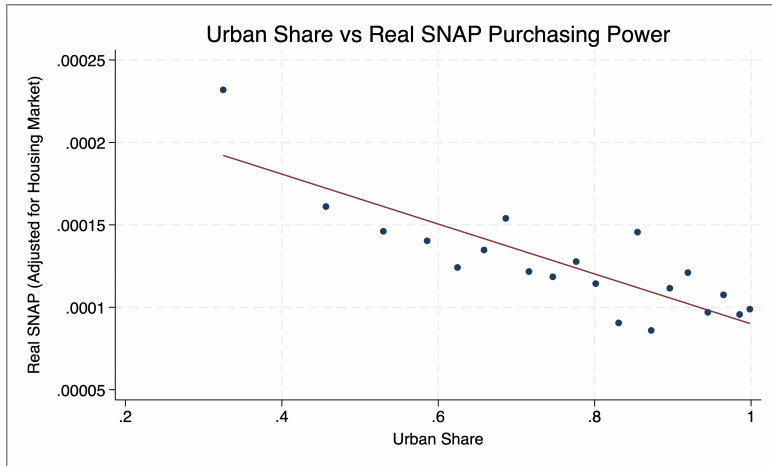
Variable	Mean	SD	Min	p25	p50	p75
Real SNAP Per Capita	0.0001266	0.0000911	0.00000501	0.0000594	0.0001085	0.0001765
Nominal SNAP	5,280,438	10,300,000	1,539,288	2,422,187	4,720,625	1.37e7
Urban Share	0.758	0.183	0.643	0.794	0.908	1.00
Percent White	0.775	0.153	0.49	0.696	0.815	0.891
Percent Black	0.126	0.131	0.031	0.083	0.181	0.72
Percent BA+	0.259	0.099	0.045	0.186	0.240	0.312
Poverty Rate	0.154	0.058	0.03	0.116	0.151	0.189
Median Income	49,934	13,388	19,083	40,828	47,028	55,216
Employment Rate	0.891	0.032	0.81	0.872	0.895	0.916
Unemployment Rate	0.108	0.032	0.02	0.085	0.106	0.128

Visualization 1: Urban Share vs Nominal SNAP



- Clear positive association.
- Driven mostly by population scale.

Visualization 2: Urban Share vs Real SNAP



- Purchasing power decreases as urban share rises.
- Reflects higher cost of living in urban counties.

$$\text{RealSNAP}_i = \alpha + \beta \text{UrbanShare}_i + \gamma' X_i + \varepsilon_i$$

- X_i : demographic + socioeconomic controls.
- β captures the **associational** relationship between urbanization and purchasing power.
- Major concern: endogeneity due to omitted cost-of-living determinants and sample restriction.

OLS Estimates of Real SNAP per Capita

	(1) Real SNAP PC	(2) Real SNAP PC (FE)
Urban Share	-0.0001514*** (0.0000259)	-0.0000407 (0.0000297)
Percent White		-0.0000829 (0.0001128)
Percent Black		-0.0000563 (0.0001139)
Percent BA Plus		-0.0003467*** (0.0000432)
Poverty Rate		0.0008332*** (0.0001218)
Median Income		0.0000000120** (0.00000000491)
Employment Rate		-0.0003795** (0.0001741)
Constant	0.0002413*** (0.0000207)	0.0001296 (0.0001895)
State FE	No	Yes
Obs.	644	571
R-squared	0.0923	0.7477

Robust SE in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

OLS Estimates of log Real SNAP per Capita

	(3) log Real SNAP PC	(4) log Real SNAP PC (FE)
Urban Share	-1.37982*** (0.1825432)	0.2798247*** (0.1074497)
Percent White		-0.257931 (0.4142353)
Percent Black		0.6532828 (0.3984258)
Percent BA Plus		-3.335995*** (0.2074816)
Poverty Rate		2.605094*** (0.4515466)
Median Income		-0.00000212*** (0.000000242)
Employment Rate		-2.223592*** (0.6509056)
Constant	-8.220868*** (0.1353132)	-8.658111*** (0.7373777)
State FE	No	Yes
Obs.	644	571
R-squared	0.0880	0.9164

Robust SE in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

- Nominal SNAP higher in urban counties → more beneficiaries.
- After adjusting for cost of living, SNAP dollars stretch less in urban areas.
- But results are **not robust**, sign-flipping after controls.
- Limited variation in urban share due to sample restricts conclusions.

Limitations

- Only counties with population $\geq 65,000 \rightarrow$ excludes rural America.
- Single cross-section (2010) $\rightarrow$ cannot use FE at county level.
- Housing value is an imperfect proxy for cost of living.
- Omitted variable bias likely substantial.

- Panel data for all counties to allow:
 - County fixed effects
 - State-by-year shocks
 - Better causal inference
- DID design comparing rural vs urban counties during recessions.
- Incorporate improved price indices (food prices, RPP, CPI local).

- SNAP benefits are uniform nationally → may disadvantage high-cost areas.
- Potential reforms:
 - Geographic benefit adjustments (similar to HUD Fair Market Rents)
 - Supplementary SNAP allotments in high-cost counties
 - Adjust benefits based on regional price parities or food-cost indices

Conclusion

- SNAP dollars do not have equal purchasing power across counties.
- Urban households face a de facto cost-of-living penalty.
- Despite data limitations, results highlight meaningful spatial disparities.
- Future work should focus on improving causal identification and cost-of-living measurement.

Thank You

Questions?